

PRISM example:

outlook	temperature	humidity	wind	play?
Sunny	hot	hi	F	No
Sunny	hot	hi	T	No
Overcast	hot	hi	F	Yes
Rainy	mild	hi	F	Yes
Rainy	cool	normal	F	Yes
Rainy	cool	normal	T	No
Overcast	cool	normal	T	Yes
Sunny	mild	hi	F	No
Sunny	cool	normal	F	Yes
Rainy	mild	normal	F	Yes
Sunny	mild	normal	T	Yes
Overcast	mild	hi	T	Yes
Overcast	hot	normal	F	Yes
Rainy	mild	hi	T	No

14
data
S
play=no.

PRISM: For each value of the class variable, call it v
create rules of the form if ? then class = v .

Example for play=no (Or, play=yes, just start over and follow same procedure)

Want if ? then play=no.

Start with single features:

outlook = S 3/5 correct

= O 0/4

= R 2/5.

temp = hot 2/4

= mild 2/6

= cool 1/4

humid = high 4/7

= normal 1/7

wind = T 3/6

= F 2/8.



choose rule with highest accuracy

i.e. outlook = S.

if tie, choose one with higher
DENOMINATOR (coverage)

But this rule not Perfect, so add more features.

PRISM.2

if outlook is overcast ? then play=no.

Choices:

temp = hot 2/2

mild 1/2

cool 0/1

humid = high 3/3 — D

= normal 0/2

wind = T 1/2

= F 2/3.

choose humidity = high. highest accuracy & tie break by denominator.

Perfect accuracy, so ~~stop with it~~ ~~stop~~ return

if ~~outlook~~ outlook = sunny & humidity = high then play=no.

Another rule for play=no?

Go until all records with play=no covered by some rule. There are 5, but only 3 covered.

Remove 3 covered records: (with outlook = sunny & humidity = high

outlook	temperature	humidity	windy	play?
overcast	hot	hi	F	yes
R	mild	hi	F	yes
R	cool	no	F	yes
R	cool	no	T	(no)
overcast	cool	no	T	yes
S	cool	no	F	yes
R	mild	no	F	yes
S	mild	no	T	yes
overcast	mild	hi	T	yes
overcast	hot	no	F	yes
R	mild	hi	T	(no)

11 left
2 nos.

want another rule if ? then play=no.

outlook = 0 0/4
 S 0/2
 R 2/5 ←

Pick outlook = R.

temp = hot 0/2
 mild 1/5
 cool 1/4

humid = hi 1/4
 normal 1/7

wind = T 2/5 ←
 F 0/6

Not perfect so continue: if outlook = R and ? then play = no

temp = hot 0/0
 mild 1/3
 cool 1/2

humid = hi 1/2
 normal 1/3

wind = T 2/2 →
 F 0/3

Perfect so stop. Rule 2: if outlook = rainy & wind = T
 then play = no.

Then start over for play = yes.